HARKUNWAR SINGH

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Data science graduate with First Class Honours MSc in Data Science and Analytics, experienced in Python, SQL, statistical modelling, machine learning and data preparation. Designed end-to-end research workflows involving historical data reconstruction, feature engineering, discrete-choice modelling and chronological evaluation, supported by hands-on software engineering and database experience.

EDUCATION

MSc Data Science and Analytics | University College Cork

Sep 2025 - Sep 2026 | *Cork, Ireland*

Classification: First Class Honours (1:1)

Relevant Coursework: Deep Learning, Data Mining, Statistical Data Analytics, Generalised Linear Modelling, Multivariate Methods, Artificial Intelligence, Information Storage & Retrieval, Relational Databases.

BTech Information Technology | Manipal University Jaipur

Oct 2020 - Oct 2024 | *Jaipur, India*

GPA: 8.7/10 | **Minor:** Data Science

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Software Engineering.

TECHNICAL EXPERIENCE

Product Engineer | GlobalLogic Technologies (Hitachi Group)

Feb 2025 - Aug 2025 | *Gurugram, India*

- Investigated enterprise application issues across Adobe Campaign, Marketo and Commerce, including access, entitlement and configuration faults.
- Analysed SQL query results and application logs to trace duplicate customer records, entitlement mismatches and data-integrity issues.
- Diagnosed SSL certificate, backend service and configuration problems; coordinated fixes with engineering and QA teams using Jira.
- Documented root causes and diagnostic steps to improve consistency across enterprise support investigations.

Engineering Trainee | HashedIn by Deloitte

Mar 2024 - Jul 2024 | *Bengaluru, India*

- Completed backend, frontend, cloud and AI development assignments during an accelerated engineering training programme.
- Developed a train reservation application using Spring Boot REST APIs, React/Redux and PostgreSQL, including seat availability, bookings and revenue reporting.
- Created a document and spreadsheet Q&A prototype with Python, FastAPI, LangChain and GPT-3.5 for ingestion, retrieval and natural-language responses.
- Containerised components with Docker, deployed training applications on AWS EC2/S3 and Azure, and configured environments with Terraform.
- Contributed to Agile stand-ups, sprint reviews and peer code reviews.

PROJECTS & RESEARCH

MSc Dissertation | Modelling Vote Transfers in Irish PR-STV Elections

Python | *R* | *Pandas* | *Conditional Logit* | *Poisson GLM* | *Random Forest* | *XGBoost*

- Engineered a research dataset from Irish general election counts (2002-2024), integrating candidate, constituency and election features; reconstructed missing structured records for 2002 from official sources and cross-checked them.
- Cleaned and reconciled historical election records, corrected data-quality issues and transformed count-level observations into choice situations for vote-transfer modelling.

- Implemented engineered features and interaction terms; compared Conditional Logit and Poisson GLM approaches with Random Forest and XGBoost baselines.
- Evaluated using random and chronological splits, holding out the 2024 election for out-of-time testing; achieved 0.0564 MAE, approximately 0.50 pseudo-R-squared and 41% top-choice accuracy against a 14.3% random baseline.

Crop Recommendation System using Fuzzy Inference

Python | *Pandas* | *NumPy* | *scikit-fuzzy* | *Flask*

- Prepared and normalised 10 years of agricultural data (2008-2017) from 33 Rajasthan districts, including rainfall, temperature, humidity and crop-yield variables.
- Designed Low/Medium/High membership functions and a Mamdani fuzzy inference engine with 27 rules per crop across five crops: Bajra, Wheat, Maize, Rice and Jowar.
- Integrated the Python/scikit-fuzzy recommendation engine into a Flask web interface for weather-based crop selection; research presented through an IEEE conference paper.

Train Reservation System

Spring Boot | *React/Redux* | *PostgreSQL*

- Implemented Spring Boot REST APIs and PostgreSQL persistence for train and passenger records, seat availability and booking workflows.

TECHNICAL SKILLS

Languages & Analytics: Python, R, SQL, Java, Pandas, NumPy, statistical modelling, GLMs, machine learning, data preprocessing

Backend & Frontend: Spring Boot, FastAPI, Flask, REST APIs, LangChain, React, Redux, HTML/CSS, Material UI

Data & Infrastructure: PostgreSQL, MySQL, AWS EC2/S3, Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Git, Jira, Linux

ADDITIONAL PROFESSIONAL EXPERIENCE

Sales Assistant | Primark (Penneys)

Feb 2026 - Present | *Cork, Ireland*

- Coordinate fitting-room operations, customer queries and store-floor workflow in a high-volume retail environment.

Sales Assistant | CeX

Nov 2025 - Feb 2026 | *Cork, Ireland*

- Maintained accurate product and inventory records in internal stock-management databases, supporting stock tracking, technology-product transactions and resolution of record discrepancies.

CERTIFICATIONS & LEADERSHIP

Google Project Management; Google IT Support; Red Hat System Administration I & II; PCAP Python; Cisco Networking Essentials; Anaconda Python for Data Science; Atlassian Agile Project Management.

Treasurer, UCC Cricket Club | Head of Promotions, Estrella Astronomy Club